

# Knowledge distillation with predicted depth for robust and lightweight face presentation attack detection

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## ABSTRACT

Face Presentation Attack Detection (FacePAD) is critical for safeguarding face recognition systems against spoofing attempts, including printed photos, video replays, and 3D masks. However, many existing approaches struggle with generalization across diverse attack types and real-world conditions. In this study, we propose a dual-branch deep learning framework that leverages both RGB images and synthetically predicted depth maps to improve anti-spoofing robustness and accuracy. A monocular depth estimation network is used to generate depth cues from a single RGB image, which are then processed in parallel with the original image through two distinct branches of a convolutional neural network. The extracted features-texture-based from RGB and structure-aware from depth-are fused via concatenation to facilitate more discriminative spoof detection. Extensive experiments on four benchmark datasets demonstrate that our method achieves state-of-the-art performance, reducing HTER to 0% on Replay-Attack and Replay-Mobile, and 1.023% on ROSE-Youtu. Similarly, an ACER of 0.56% is achieved on OULU-NPU, while maintaining computational efficiency. Furthermore, we introduce a knowledge distillation scheme to compress the dual-branch model into a lightweight single-branch variant suitable for real-time deployment in mobile authentication, surveillance, and biometric access control scenarios.

## 1. Introduction

Face recognition systems play a vital role in numerous computer vision and interactive AI applications. They are widely used in everyday scenarios such as smartphone unlocking [1], mobile payments [2], access control [3], and CCTV surveillance [4]. The quality of face representations has been significantly enhanced through multi-branch and fusion-based deep learning techniques, which effectively integrate global structure and fine-grained texture cues [5]. Further improvements have been achieved by adaptive fusion strategies that compensate for data compression and degradation effects [6], reinforcing the benefits of parallel processing pipelines in facial analysis. Despite their widespread adoption, these systems are vulnerable to various face presentation attacks, including print attacks (e.g., printed photos), video replay attacks (e.g., replaying face videos on screens), and 2D/3D mask attacks. These vulnerabilities can significantly undermine the reliability of face recognition systems in real-world applications. Consequently,

face anti-spoofing, also known as face presentation attack detection (FacePAD), has emerged as a critical task for determining whether an input face is live or spoofed.

To counteract presentation attacks, several FacePAD methods [7–10] have been proposed. These techniques can be broadly categorized into two major paradigms: traditional handcrafted-feature-based approaches and deep learning-based methods. Traditional approaches extract texture-based features such as LBP [11], SIFT [12], and SURF [13], and classify them using machine learning algorithms like LDA [14] or SVM [15]. However, these methods often fail against high-quality replay or 2D/3D mask attacks due to their limited capacity to generalize.

In contrast, CNN-based approaches have shown promising results by learning discriminative features in an end-to-end manner. Recent advances in semantic segmentation [16] and image enhancement [17,18] tasks further demonstrate how end-to-end CNNs can effectively model contextual relationships and adapt to degraded inputs. Many image-based FacePAD techniques [8–10,19–21] treat spoof detection as a

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binary classification task; classifying faces as either bonafide or attack [22]. However, these models often suffer from domain generalization issues. When trained on specific attack patterns or environmental conditions, the models tend to learn unfaithful cues such as screen bezels or background artifacts, leading to poor generalization across datasets.

To mitigate this problem, Liu et al. [23] introduced remote photoplethysmography (rPPG) signals as auxiliary supervision. While rPPG-based methods provide valuable physiological cues to distinguish live from spoofed faces, they are computationally intensive and unsuitable for real-time applications. To overcome latency issues, depth-supervised approaches have been proposed, which incorporate auxiliary sensors to estimate facial depth. For example, Atoum et al. [24] and Liu et al. [23] used depth signals to guide CNNs in learning more generalizable features. Since bonafide faces contain natural depth variations while spoofed faces typically appear planar, depth-maps offer effective discriminative information, leading to real-time performance and improved robustness.

However, relying on external depth sensors introduces deployment challenges, particularly for cost-sensitive or mobile environments where such hardware may not be available. This limitation necessitates alternative strategies for inferring depth information without additional sensors.

To address these challenges, we propose a novel FacePAD framework that estimates structural and depth-aware features directly from RGB images. Leveraging recent advances in monocular depth estimation (e.g., Depth Anything [25,26]), we synthesize depth-maps from single RGB inputs and design a *dual-branch deep learning architecture* that processes both RGB and estimated depth images in parallel. The RGB branch captures fine-grained texture cues, while the depth branch extracts structural information, enabling more robust detection of sophisticated spoofing attacks. Feature representations from both branches are fused before classification to exploit complementary information.

While dual-branch models enhance detection accuracy, they are typically computationally expensive. To reduce inference cost, we introduce a knowledge distillation [27] framework that transfers knowledge from the dual-branch architecture (teacher) to a lightweight single-branch MobileNetV3-based [28] student model. Our experiments show that the student model achieves competitive performance with significantly reduced complexity, making it suitable for real-time deployment.

The main contributions of this paper are summarized as follows:

- A dual-branch deep learning architecture is proposed that integrates RGB images and predicted depth-maps to enhance FacePAD performance.
- A novel integration of deep learning-based monocular depth estimation model is proposed to generate synthetic depth cues, improving resilience against advanced 3D mask and replay attacks.
- A knowledge distillation framework is designed to remove the dependency on the depth estimation network (DEN) during inference, significantly reducing computational overhead.

The rest of this paper is organized as follows: [Section 2](#) reviews related works in FacePAD. [Section 3](#) details the proposed methodology and model architecture. [Section 4](#) presents experimental results and analysis. Finally, [Section 5](#) concludes the paper and outlines future research directions.

## 2. Background and related work

This section reviews the evolution of FacePAD techniques and categorizes existing literature into three main groups: (1) Deep learning-based binary supervised methods, (2) Temporal information-based methods, and (3) Depth-supervised methods.

### 2.1. Deep learning-based binary supervised methods

Face anti-spoofing has traditionally been approached as a binary classification task, distinguishing between live and spoofed facial inputs. Early approaches relied on hand-crafted features such as LBP [29–31], SIFT [32], SURF [33], HOG [34,35], and DoG [36,37], which were then classified using SVM or LDA models. However, these manually engineered features are highly sensitive to variations in environmental conditions such as lighting, camera type, and the nature of the presentation attack instruments, leading to limited generalization across datasets.

With advances in computational power and the availability of large-scale datasets, Convolutional Neural Networks (CNNs) have become a dominant paradigm for FacePAD [8–10,20,22,38,39]. These methods learn discriminative representations in an end-to-end fashion [16], often supervised using softmax loss in a binary classification setup. For instance, Li et al. [22] and Patel et al. [39] fine-tuned pre-trained VGG-Face networks for spoof detection, while Nagpal et al. [10] systematically evaluated the influence of network architecture and training settings. Similarly, Feng [38] and Li [22] trained CNNs on multi-view facial inputs to enhance spoofing detection performance. Despite their success, binary-supervised CNN methods remain vulnerable to overfitting to dataset-specific artifacts (e.g., screen bezels, lighting), leading to weak domain generalization.

### 2.2. Temporal information-based methods

Temporal dynamics provide essential cues for distinguishing live faces from spoofed ones. Early works such as [39,40] exploited eye-blinking patterns to detect spoofing attempts. While useful, these methods were limited in scope, especially against high-quality replay attacks, which may simulate similar behaviors.

To address this, recent approaches have incorporated temporal modeling to capture motion patterns across multiple frames. For example, Gan et al. [8] proposed a 3D CNN that learns spatio-temporal features from video input. However, the performance of 3D CNNs often deteriorates on small datasets due to their high capacity and data-hungry nature. Xu et al. [19] combined CNNs with LSTMs to model temporal dependencies in video sequences, treating spoof detection as a sequence learning problem. Feng et al. [38] used a fusion of optical flow magnitude maps and Shearlet features to capture motion inconsistencies indicative of spoofing. These optical flow cues helped differentiate subtle facial movements in bonafide videos from static or repetitive patterns in spoof attacks.

More recently, Khan et al. [41] introduced a spatio-temporal averaging strategy to balance performance and computational complexity by leveraging averaged temporal features without relying on expensive 3D convolutions or LSTMs. Despite such advancements, many temporal-based methods still struggle to model long-range dependencies and are sensitive to video quality, frame rate, and motion artifacts.

### 2.3. Depth-supervised methods

Compared to binary-supervised approaches, depth-supervised methods leverage geometric cues to enhance robustness and generalization. Atoum et al. [24] introduced a two-stream architecture that combines patch-based and depth-based CNNs. Their method extracted both local texture and holistic depth information from facial images to improve spoof detection. The use of estimated depth helped distinguish between planar spoof mediums (e.g., print, screen) and the 3D structure of live faces.

Building upon this, Liu et al. [23] incorporated both depth and remote photoplethysmography (rPPG) signals into a unified framework. Their model employed a recurrent neural network (RNN) to process temporal rPPG features and a non-rigid registration layer to align facial regions under varying poses and expressions. Although effective, their

normalization mechanism risked suppressing spoof-specific anomalies such as unnatural facial motions or abrupt transitions.

Overall, depth-supervised methods have shown great promise in capturing structural differences between real and spoofed faces. However, most rely on either a dedicated depth sensors or a compute extensive DEN, which limit their scalability and practical deployment in consumer-grade devices lacking 3D imaging or heavy compute capabilities.

### 3. Materials and methods

This section describes the datasets used, performance evaluation metrics, the proposed network architecture, and the training methodology employed in this study.

#### 3.1. Datasets

To evaluate the effectiveness of the proposed method, four widely-used face anti-spoofing datasets are utilized: Replay-Attack [42], Replay-Mobile [43], ROSE-Youtu [44], and OULU-NPU [45]. The details of each dataset are listed in Table 1.

##### 3.1.1. Replay-attack dataset

The Replay-Attack dataset [42] contains 1300 short video clips that include both genuine face presentations and spoofing attacks (photo and video replays). It consists of 50 subjects recorded under controlled and adverse lighting conditions using various cameras. The dataset is divided into training, development, and test sets, ensuring that subjects do not overlap between sets. Fig. 1(a) shows example frames from real and spoofed videos.

##### 3.1.2. Replay-mobile dataset

The Replay-Mobile dataset [43] is designed specifically for mobile face recognition and FacePAD. It comprises 1164 video clips from 39 subjects, following the recent removal of one subject's data at the data manager's request (originally 1190 video clips from 40 subjects). These video clips are captured under various lighting conditions using mobile devices. The dataset includes both printed photo and digital replay attacks. Fig. 1(b) illustrates real and attack frames from this dataset.

##### 3.1.3. ROSE-Youtu dataset

The ROSE-Youtu dataset [44] contains 3350 video clips from 20 subjects, recorded under diverse lighting conditions and using multiple camera models. It includes three attack types: video replay, print photo,

and paper mask attacks. Fig. 1(c) shows example frames from real and spoofed videos in the dataset. The dataset provides a challenging benchmark for face anti-spoofing research.

##### 3.1.4. OULU-NPU dataset

The OULU-NPU dataset [45] is a high-resolution face anti-spoofing dataset designed to simulate real-world scenarios with various conditions. It includes 4950 video recordings collected from 55 subjects using six different illumination settings and three different background scenes. Attacks in this dataset are based on printed photos and video replays, captured using two mobile phones. The dataset is also arranged into four distinct protocols to facilitate robust evaluation. Each protocol targets a specific generalization aspect such as cross-illumination, cross-device, and cross-environment performance. Fig. 1(d) illustrates sample real and attack frames from this dataset.

#### 3.2. Data pre-processing and augmentation

We employ a systematic data pre-processing and augmentation pipeline (see Fig. 2). The input data consists of synchronized pairs of RGB videos and their corresponding predicted depth videos, or in some cases only RGB videos when depth-maps are not utilized.

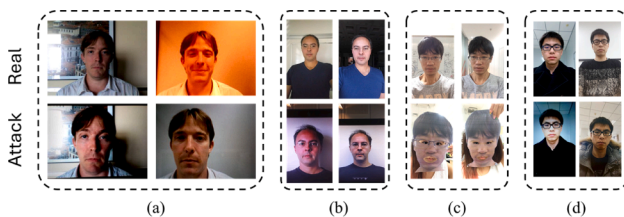
In each training epoch iteration, we randomly sample a single frame from each video clip to introduce variability and improve model generalization across different facial expressions, poses, and lighting conditions. Each selected frame undergoes a standardized pre-processing sequence to ensure uniformity across all samples. The largest possible centered square region is cropped from each frame for adaptive center cropping. This ensures that the face region remains the primary focus while minimizing background content. The cropped square image is then resized to a resolution of  $224 \times 224$  pixels using Lanczos resampling, preserving fine spatial details essential for spoof detection, as illustrated in Fig. 2(a).

To further promote robustness and prevent overfitting during training, few random augmentations are also applied with 50% probability. Where frames are rotated by an angle randomly sampled from the range  $[-180^\circ, 180^\circ]$ , and are uniformly scaled with a random factor between 0.7 and 1.3. An illustration for these augmentations has been provided in Fig. 2(b).

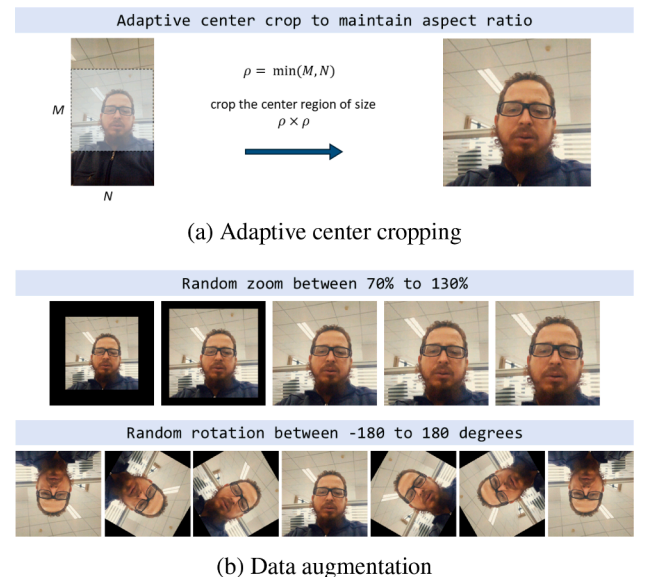
For experiments involving depth-augmented hybrid models, corresponding predicted depth frames are processed using identical transformation functions. Depth frames are first converted to single-channel grayscale before adaptive cropping and resizing. And identical

**Table 1**  
Key attributes of the datasets utilized in this study.

| Dataset            | Year | Subjects | Genuine/Spoof Clips |
|--------------------|------|----------|---------------------|
| Replay-Attack [42] | 2012 | 50       | 300 / 1000          |
| Replay-Mobile [43] | 2016 | 39       | 540 / 624           |
| ROSE-Youtu [44]    | 2018 | 20       | 1000 / 2350         |
| OULU-NPU [45]      | 2017 | 55       | 990 / 3960          |

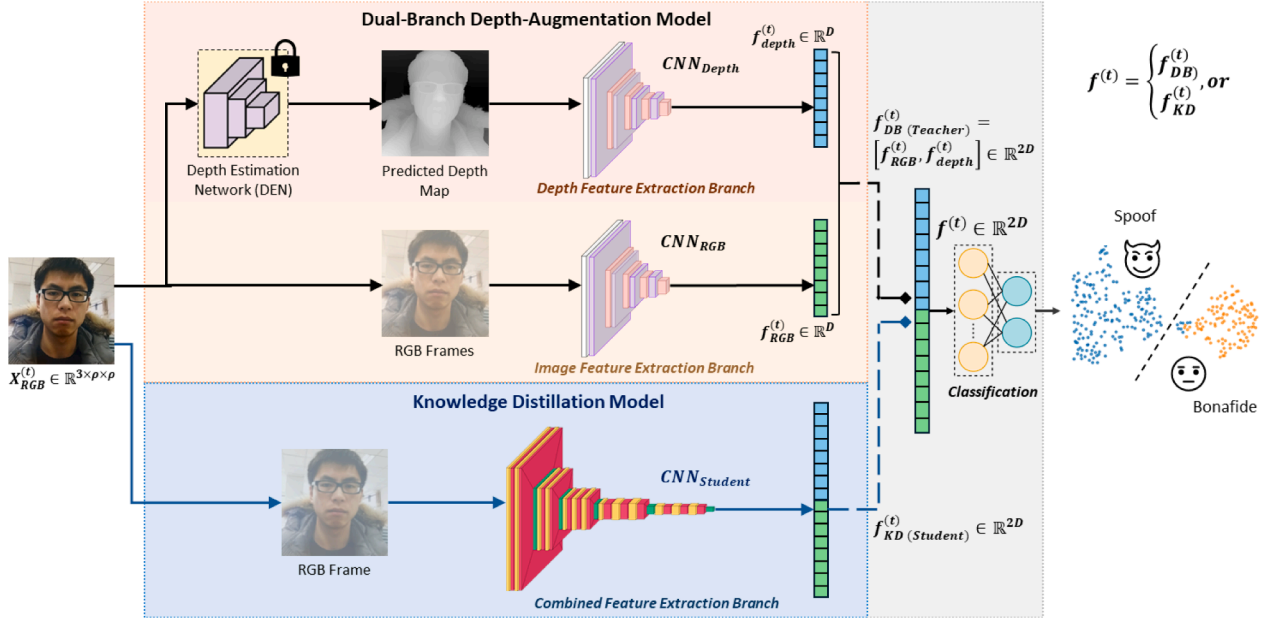


**Fig. 1.** Example of real and attack video frames in the (a) Replay-Attack, (b) Replay-Mobile, (c) ROSE-Youtu, and (d) OULU-NPU datasets. The top row represents real frames, while the bottom row presents attack frames using various attack instruments.



**Fig. 2.** Data pre-processing and augmentation methods used in this study.

## Knowledge Distillation with Predicted Depth



**Fig. 3.** Illustration of the proposed Dual-Branch depth-augmentation, and Knowledge Distillation architectures for FacePAD. In the Dual-Branch model, RGB frames are processed through two parallel branches for image and depth features extraction. The depth branch utilizes synthetic depth maps predicted by the frozen DEN (indicated by the lock symbol). Features from these two branches are explicitly fused via concatenation to form a joint representation  $\mathbf{f}^{(t)}$ , subsequently classified into spoof or bonafide categories using fully connected layers. For efficient real-time deployment, knowledge from the Dual-Branch model as a Teacher is distilled into a lightweight single-branch Student model, which directly extracts the combined modality-aware representation  $\mathbf{f}^{(t)}$  from RGB inputs alone.

augmentation parameters are applied to both modalities to preserve alignment between texture and depth information.

### 3.3. Depth estimation network

We adopt the *Depth Anything V2* model [26] to extract depth cues critical for our depth-augmented FacePAD models. This state-of-the-art monocular depth estimation network utilizes a transformer-based encoder-decoder architecture, with a ViT-G backbone in the teacher model and smaller ViT-S/B/L variants for student networks [46,47]. The teacher is trained on 595K synthetic RGB-depth pairs and then used to pseudo-label 62 million real-world RGB images, which serve as training data for student models via supervised depth regression.

In this study, we employ the ViT-B variant (Depth-Anything-V2-Base), which offers an optimal balance between generalization and computational efficiency. This model is used consistently across all experiments unless otherwise stated.

Depth map inference is performed offline for all training and evaluation data. Each extracted RGB frame is passed through the pretrained and frozen DEN to generate a corresponding synthetic depth map. These depth maps are treated as primary inputs for the depth branch of our proposed Dual-Branch architecture (see Fig. 3), allowing for tightly aligned texture and depth information.

While the depth estimation is decoupled from end-to-end training for computational efficiency, it is retained as a frozen module in our practical deployment pipeline. In real-time scenarios, the DEN infers depth maps on-the-fly from each RGB frame. Furthermore, for deployment in resource-constrained settings, we also explore a distilled RGB-only variant that removes the depth branch entirely (see Section 3.4.2).

### 3.4. Proposed method and network architecture

Given a video sequence  $\{V^{(t)}\}_{t=1}^T$ , where each RGB frame  $V^{(t)} \in \mathbb{R}^{3 \times H \times W}$ , the goal is to classify each frame as either *genuine* or *spoofed*. Our proposed framework utilizes synthetic depth-maps  $D^{(t)} \in \mathbb{R}^{1 \times H \times W}$

generated by the monocular depth estimation model *Depth Anything V2* [26]. Both RGB and depth frames are independently processed by dedicated CNN branches to extract modality-specific  $D$ -dimensional feature vectors from each modality:

$$\mathbf{f}_{RGB}^{(t)} = \text{CNN}_{RGB}(V^{(t)}) \in \mathbb{R}^D, \quad (1)$$

$$\mathbf{f}_{Depth}^{(t)} = \text{CNN}_{Depth}(D^{(t)}) \in \mathbb{R}^D. \quad (2)$$

These modality-specific features are then concatenated into a fused representation:

$$\mathbf{f}^{(t)} = [\mathbf{f}_{RGB}^{(t)}, \mathbf{f}_{Depth}^{(t)}] \in \mathbb{R}^{2D}. \quad (3)$$

The fused vector  $\mathbf{f}^{(t)}$  is fed into a fully connected (FC) layer to predict the class scores:

$$s(c|\mathbf{f}^{(t)}) = W_{FC}\mathbf{f}^{(t)} + \mathbf{b}_{FC} \in \mathbb{R}^2, \quad (4)$$

where  $c \in \{\text{genuine, spoofed}\}$  denotes the class label.

#### 3.4.1. Implementation setup for dual-branch FacePAD

Table 2 summarizes key architectural details, and the overall pipeline can be described as follows:

**Table 2**

Overview of the proposed dual-branch FacePAD architecture with RGB and depth modality-specific feature extraction.

| Layer Type         | RGB Branch                           | Depth Branch                         | Output Shape        |
|--------------------|--------------------------------------|--------------------------------------|---------------------|
| Input              | $B \times 3 \times \rho \times \rho$ | $B \times 1 \times \rho \times \rho$ | —                   |
| Feature Extraction | $\text{CNN}_{RGB}$                   | $\text{CNN}_{Depth}$                 | $B \times 960$ each |
| Feature Fusion     | Concatenation                        |                                      | $B \times 1920$     |
| Classifier         | Two-layer MLP                        |                                      | $B \times 2$        |

*Notation:*  $B$  = batch size;  $\rho$  = input resolution height or width; each RGB image has 3 channels, and each predicted depth-map has 1 channel.

- **Frame Acquisition:** A single RGB frame is randomly sampled from each video sequence, then adaptively cropped and resized to a resolution of  $\rho \times \rho$ .
- **Depth Prediction:** A corresponding synthetic depth-map is generated for each RGB frame, processed similarly to the RGB frame to ensure spatial alignment.
- **CNN Backbones:** We adopt MobileNetV3-Large as the backbone for both the RGB and depth branches due to its excellent trade-off between efficiency and representational power, particularly suited for real-time and resource-constrained FacePAD applications [41,48,49]. Both networks are initialized with ImageNet1K\_V2 pretrained weights [28,50]. To accommodate the single-channel input of synthetic depth maps, the first convolutional layer of the depth branch is modified accordingly. Each branch independently extracts a  $D = 960$  dimensional feature vector from its respective modality. This dual-branch setup enables the model to capture complementary information from appearance and geometric cues.
- **Feature Fusion:** The dual-branch features from the RGB and depth modalities CNN backbone branches are fused explicitly by concatenation to form a joint feature vector  $\mathbf{f}^{(i)} \in \mathbb{R}^{2D=1920}$  (illustrated in Fig. 3).
- **Classifier:** A two-layer MLP performs binary classification, distinguishing genuine from spoofed faces.

Fig. 3 illustrates the complete Dual-Branch (DB) architecture, highlighting separate processing of RGB and depth modalities, feature fusion via concatenation, and classification. The detailed implementation steps are formally summarized in Algorithm 1.

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**Algorithm 1: Dual-Branch FacePAD Model with Depth Prediction**


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**Input:** Batch of video sequences  
 $V \in \mathbb{R}^{B \times r \times 3 \times H \times W}$

**Output:** Class predictions  $Y \in \mathbb{R}^{B \times 2}$

**Given:**  
 DepthEstimator: pretrained monocular depth estimator;  
 $\text{CNN}_{\text{RGB}}$  : trained RGB feature extractor;  
 $\text{CNN}_{\text{Depth}}$  : trained depth feature extractor;  
 FC: trained fully connected classifier.

```

1 Initialize empty list  $Y = []$ 
2 for each video  $V^{(i)}$  in batch do
3   Frame Selection:
4   Sample RGB frame  $X_{\text{RGB}}^{(i)} \in \mathbb{R}^{3 \times \rho \times \rho}$ .
5   Depth Prediction:
6   Estimate synthetic depth-map:
7    $X_{\text{Depth}}^{(i)} = \text{DepthEstimator}(X_{\text{RGB}}^{(i)}) \in \mathbb{R}^{1 \times \rho \times \rho}$ .
8   Feature Extraction:
9   RGB features:  $\mathbf{f}_{\text{RGB}}^{(i)} = \text{CNN}_{\text{RGB}}(X_{\text{RGB}}^{(i)}) \in \mathbb{R}^D$ .
10  Depth features:  $\mathbf{f}_{\text{Depth}}^{(i)} = \text{CNN}_{\text{Depth}}(X_{\text{Depth}}^{(i)}) \in \mathbb{R}^D$ .
11  Feature Fusion:
12  Concatenate modality-specific features:
13   $\mathbf{f}^{(i)} = [\mathbf{f}_{\text{RGB}}^{(i)}; \mathbf{f}_{\text{Depth}}^{(i)}] \in \mathbb{R}^{2D}$ .
14  Classification:
15  Predict class scores:  $y^{(i)} = \text{FC}(\mathbf{f}^{(i)}) \in \mathbb{R}^2$ .
16  Append  $y^{(i)}$  to  $Y$ .
17 end
18 return Class predictions  $Y$ 

```

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The model is trained using the *Adam optimizer* with an initial learning rate of  $10^{-4}$ . The objective function is the cross-entropy loss, defined as:

$$L = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^2 y_{i,c} \log(p_{i,c})$$

where  $y_{i,c}$  is a binary indicator (0 or 1) denoting whether class label  $c$  is the correct classification for sample  $i$ , and  $p_{i,c}$  is the predicted probability for class  $c$  obtained after applying a softmax function to the network outputs. Training is conducted using batch size of 16 for a maximum of 100 epochs, with an early stopping patience of 10 consecutive epochs contingent on validation loss, and testing is performed with a batch size of 256.

### 3.4.2. Implementation setup for knowledge distillation model

The objective of the knowledge distillation (KD) stage is to eliminate the need for explicit depth estimation during inference, thereby reducing computational overhead while retaining the performance benefits of dual-modality learning. To achieve this, we employ a single-branch student network, denoted by  $\text{CNN}_{\text{Student}}(X_{\text{RGB}}^{(i)})$  as illustrated in Fig. 3, which is trained to mimic the feature representations and predictions of the full Dual-Branch (teacher) model.

In the KD setup, the teacher model consists of the complete Dual-Branch pipeline, comprising the Depth Anything V2 network for depth-map prediction and two MobileNetV3-Large backbones for separate feature extraction from RGB and depth inputs. In contrast, the student model is a single MobileNetV3-Large network that processes only the RGB input. The student's architecture is adapted so that its final feature extraction layer outputs a vector of size  $2D$ , where  $D$  is the dimensionality of features in each teacher branch. This design enables the student to approximate the fused representation  $\mathbf{f}^{(i)} = [\mathbf{f}_{\text{RGB}}^{(i)}; \mathbf{f}_{\text{Depth}}^{(i)}] \in \mathbb{R}^{2D}$  directly from the input image, without requiring a depth-map as input.

During training, the KD framework leverages both the standard cross-entropy loss with ground-truth labels and a logit-based distillation from a frozen teacher network. Specifically, we employ a combination of the following losses:

- **Classification Loss:** Cross-entropy loss  $L_{\text{CE}}$  between the student's predictions and the true class labels.
- **Distillation Loss:** Kullback-Leibler (KL) divergence  $L_{\text{KL}}$  [27,51] between the softened output distributions of the teacher and student, computed at temperature  $T$ .

The overall training objective can be written as:

$$L_{\text{KD}} = (1 - \alpha) \cdot L_{\text{CE}} + \alpha \cdot T^2 \cdot L_{\text{KL}} \quad (5)$$

where  $\alpha$  controls the relative weight of the distillation loss, and the temperature  $T$  smooths the predicted logits to improve gradient flow during training.

The student network is trained using the Adam optimizer with an initial learning rate of  $10^{-4}$ . Training is performed with a batch size of 16 for up to 100 epochs, with early stopping applied based on validation loss. At inference time, only the lightweight single-branch student model is used, which requires only RGB images and eliminates the computationally expensive depth estimation module.

This knowledge distillation approach allows the final deployed model to maintain much of the performance and robustness of the Dual-Branch teacher, while being highly efficient and suitable for real-time and resource-constrained environments.

### 3.5. Performance evaluation metrics

The proposed system is evaluated using both computational and classification metrics. Classification metrics include test accuracy, area under the receiver operating characteristic curve (AUC-ROC), and half total error rate (HTER) across all datasets. For the OULU-NPU dataset, additional PAD metrics including attack presentation classification error rate (APCER), bonafide presentation classification error rate (BPCER), and average classification error rate (ACER) are also evaluated. The HTER is calculated as follows:

$$\text{HTER} = \frac{\text{FAR} + \text{FRR}}{2}$$

where FAR and FRR are the false acceptance rate and false rejection rate defined by:

$$FAR = \frac{FP}{FP + TN}, \quad FRR = \frac{FN}{FN + TP}$$

And,  $TP$ ,  $TN$ ,  $FP$ , and  $FN$  represent the number of true positives, true negatives, false positives, and false negatives, respectively.

The AUC-ROC is computed by plotting the receiver operating characteristic curve across different thresholds and calculating the area under this curve, providing a threshold-independent measure of classification performance. The equal error rate (EER) is the point on the ROC curve where the FAR equals the FRR, providing a single-threshold summary of classifier performance. The Youden Index (Max) is defined as the maximum difference between the TPR and FPR across all thresholds (*Youden Index (Max)* =  $\max(TPR - FPR)$ ). It indicates the optimal threshold that balances sensitivity and specificity.

For the OULU-NPU dataset, APCER measures the proportion of attack samples incorrectly classified as bonafide, corresponding to the attack-specific false acceptance rate. BPCER quantifies the proportion of bonafide samples incorrectly classified as attacks, corresponding to the bonafide-specific false rejection rate. ACER is then calculated as the mean of APCER and BPCER, defined as:

$$ACER = \frac{APCER + BPCER}{2}$$

These metrics collectively provide a comprehensive evaluation of the proposed FacePAD system's effectiveness in detecting presentation attacks.

## 4. Results and discussion

This section presents a comprehensive performance analysis of the proposed FacePAD method evaluated across several benchmark datasets: Replay-Attack (RA), Replay-Mobile (RM), ROSE-Youtu (RY), and OULU-NPU. We provide detailed insights into accuracy, error rates, and computational efficiency, followed by comparisons with current state-of-the-art (SOTA) FacePAD methods.

### 4.1. Performance evaluation of image- and depth-based models

Table 3 summarizes key performance metrics, including HTER, AUC-ROC, Accuracy, APCER, BPCER, and ACER; for image-only, depth-only, and proposed dual-branch models across all datasets.

Results indicate that the proposed Dual-Branch architecture consistently outperforms single-modality (RGB or depth-only) models across all datasets. Particularly, the Dual-Branch model achieves an HTER of 0.313 % and an accuracy of 99.610 % on the OULU-NPU dataset, significantly exceeding performance observed in single-modality evaluations.

Fig. 4 illustrates t-SNE plots for embeddings from RGB, depth, and fused Dual-Branch model. The dual-branch embeddings show superior class separability, confirming the advantage of jointly leveraging RGB and depth modalities.

Furthermore, Grad-CAM visualizations presented in Fig. 5 demonstrate that the Dual-Branch model accurately localizes regions of interest essential for differentiating between genuine and spoofed faces, indicating effective spatial attention and interpretability.

### 4.2. Ablation study

An ablation study was conducted on the ROSE-Youtu dataset to analyze the influence of depth-augmentation methods, transfer learning strategies, DEN variant for depth-map quality, and network architectures.

#### 4.2.1. Effect of different depth-augmentation techniques

We evaluated five depth-augmentation strategies: (1) a multi-channel input model, (2) an auxiliary depth input model, (3) an independent image and depth feature extraction dual branch model with image-depth feature fusion via summation, (4) an independent image and

**Table 3**

Results on RA, RM, RY, and OULU datasets for image only, depth only, and dual-branch models.

| Datasets Model | Metric   | Image-based | Depth-map-based | Dual-Branch |
|----------------|----------|-------------|-----------------|-------------|
| RA             | HTER     | 0           | 10.620          | 0           |
|                | AUC-ROC  | 1           | 0.960           | 1           |
|                | Accuracy | 100         | 91.250          | 100         |
| RM             | HTER     | 0           | 3.900           | 0           |
|                | AUC-ROC  | 1           | 0.995           | 1           |
|                | Accuracy | 100         | 96.030          | 100         |
| RY             | HTER     | 1.472       | 9.000           | 1.023       |
|                | AUC-ROC  | 0.999       | 0.969           | 0.999       |
|                | Accuracy | 98.055      | 92.220          | 99.027      |
| OULU-NPU       | HTER     | 0.625       | 13.299          | 0.313       |
|                | AUC-ROC  | 1           | 0.930           | 1           |
|                | Accuracy | 98.890      | 86.940          | 99.610      |
|                | APCER    | 1.944       | 12.222          | 0.278       |
|                | BPCER    | 0.556       | 21.389          | 1.111       |
|                | ACER     | 1.250       | 16.806          | 0.694       |

depth feature extraction dual branch model with image-depth feature fusion via attention weights, and (5) the proposed *Dual-Branch* model with separate RGB and depth streams, with extracted image-depth features fusion via concatenation, prior to classification.

In the Multi-Channel input approach, RGB and predicted depth frames are concatenated to form a 4-channel input, introducing *Channel Expansion* to accommodate both modalities (see Fig. 6(a)). This expanded input is processed by a modified MobileNetV3-Large model with its first convolutional layer adapted for 4 channels, where the additional depth input channel weights are initialized as the mean of the RGB channels weights.

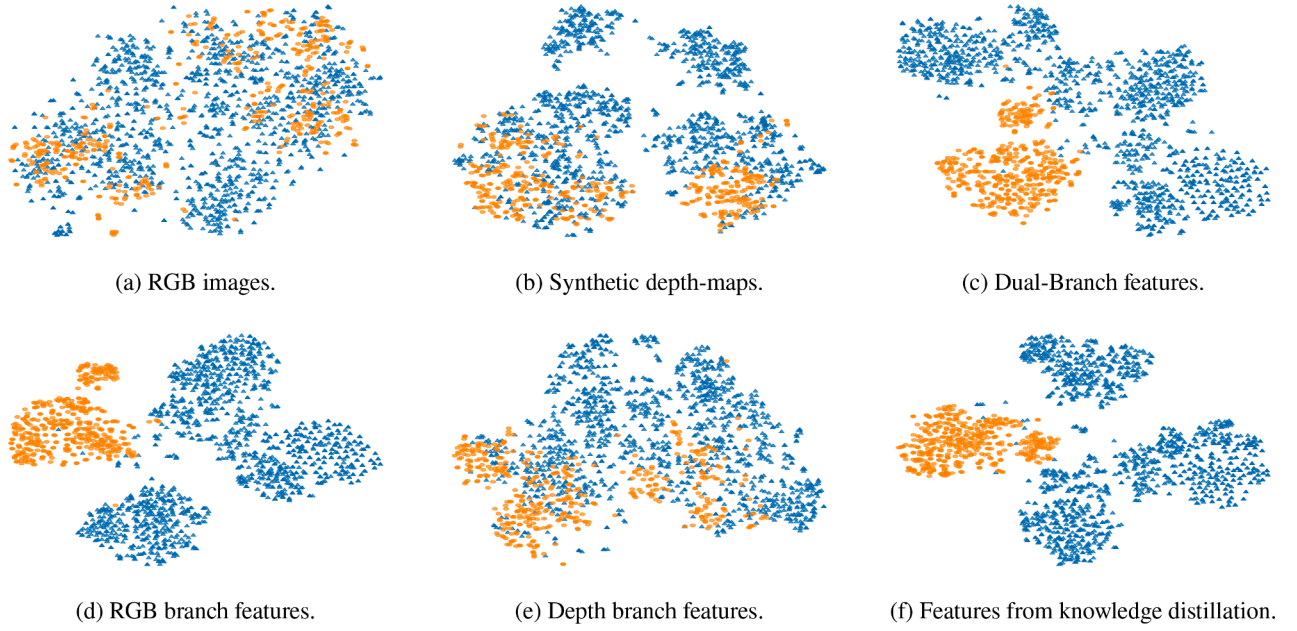
In the *Auxiliary Depth* input approach, the standard MobileNetV3-Large model backbone is utilized for RGB feature extraction, and a lightweight CNN is used to extract depth features (see Fig. 6(b)). This auxiliary CNN consists of two convolutional layers with ReLU activations and max pooling, followed by global average pooling to produce a compact depth feature vector. The resulting vector is projected to match the dimensionality of the RGB backbone features through a fully connected layer with a sigmoid activation, generating learned scaling weights. These weights are then applied element-wise to modulate the RGB feature vector prior to final classification, allowing the depth signal to refine spatial representations without modifying the backbone.

In the *Dual-Branch* models, RGB and depth frames are processed independently using two MobileNetV3-Large backbones. The RGB branch handles the standard 3-channel input, while the depth branch accepts single-channel input via a modified first layer. Each branch outputs a 960-dimensional feature vector, and these are fused before classification. We evaluate three fusion strategies: summation, attention, and concatenation-referred to as *Dual-Branch (Summation)*, *Dual-Branch (Attention)*, and *Dual-Branch (Concatenation)* models.

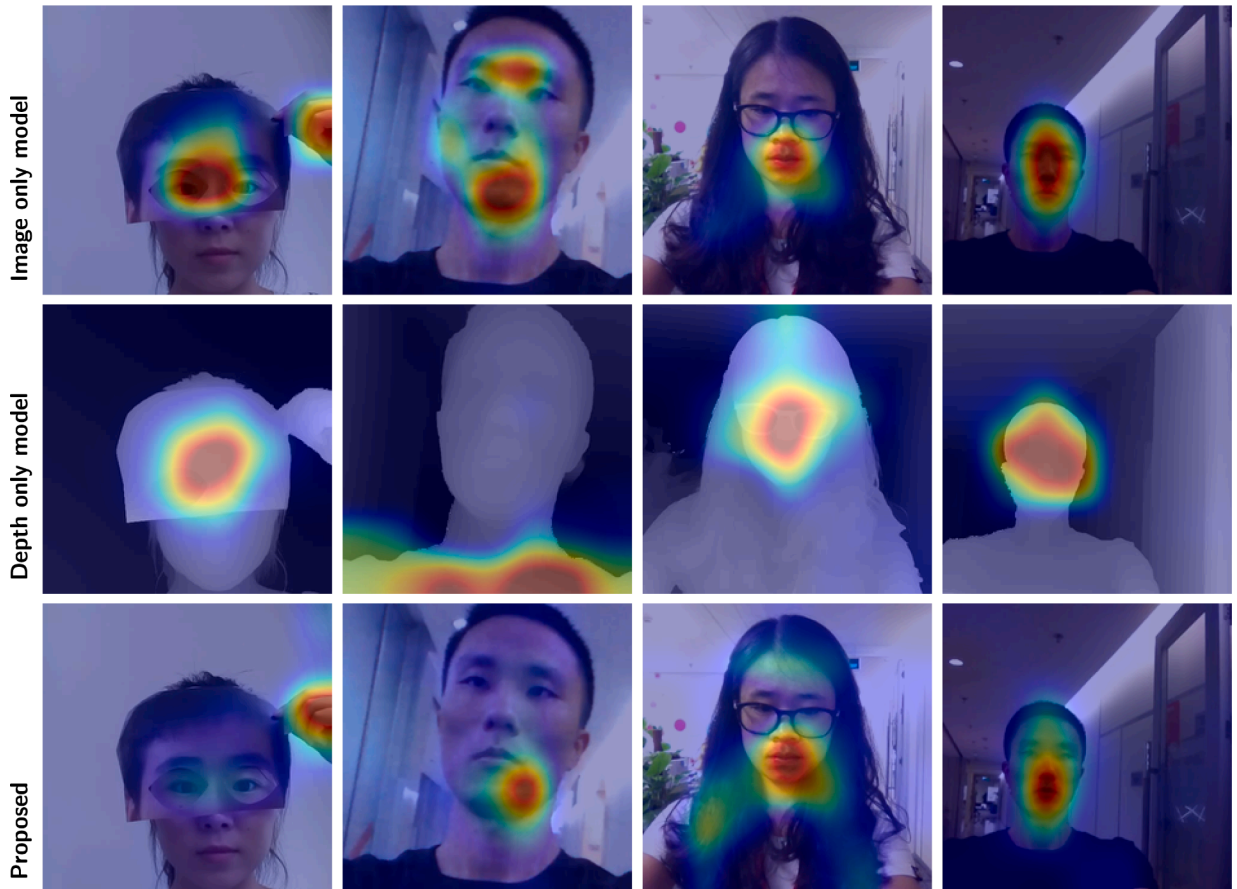
In the *Dual-Branch (Summation)* model, features extracted from the RGB and depth branches are combined via element-wise summation (see Fig. 6(c)), offering a compact and efficient fusion mechanism that emphasizes shared modality representations.

In the *Dual-Branch (Attention)* model, concatenated features are passed through a squeeze-and-excitation-style attention module that learns channel-wise weights to modulate the relative influence of RGB and depth cues via a weighted sum (see Fig. 6(d)). This allows adaptive feature fusion depending on input characteristics.

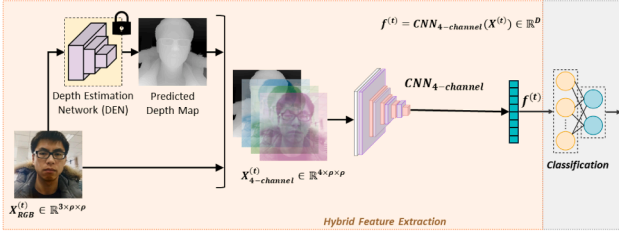
Lastly, the *Dual-Branch (Concatenation)* model, discussed in detail in Section 3.4.1 and used as the default configuration (referred to as the *DB model*), concatenates the two feature vectors and feeds the combined representation to the classifier (Fig. 3).



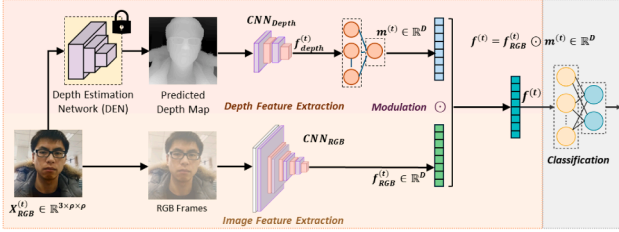
**Fig. 4.** t-SNE visualizations showing feature separability for FacePAD. Each point represents one sample, colored by class (real:Orange or spoofed:Blue). The Dual-Branch architecture (c) achieves superior discriminative power compared to single modalities (a, b) and isolated branches (d, e). The distilled model (f) maintains effective discrimination suitable for real-time deployment.



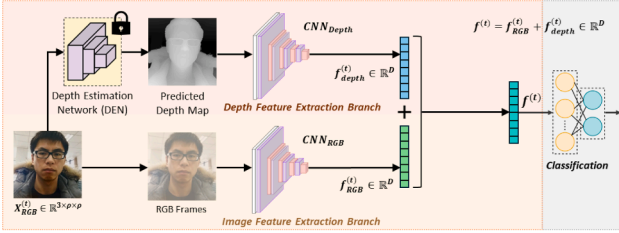
**Fig. 5.** Grad-CAM visualizations for image only, depth only, and proposed KD models (rows) on multiple samples (columns). The first two columns are attack samples, and the last two are real samples.



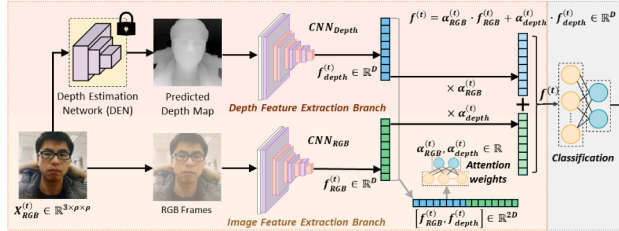
(a) Channel expansion model architecture with 4-channels of (R, G, B, Depth-map).



(b) Auxiliary depth input model architecture.



(c) Dual-Branch (Summation) model architecture.



(d) Dual-Branch (Attention) model architecture.

Fig. 6. Model architectures for depth-map augmentation techniques tested in this study.

Quantitative results in Table 4 show that the Dual-Branch (Concatenation) model consistently outperforms the others, achieving the lowest HTER (0.31 %) and ACER (0.69 %) on OULU-NPU and the highest accuracy (99.03 %) on ROSE-Youtu.

#### 4.2.2. Effect of transfer learning

We analyzed the effect of pretrained versus non-pretrained MobileNetV3-Large backbones on the ROSE-Youtu dataset. Results in Table 5 demonstrate a significant benefit of transfer learning, substantially reducing HTER from 11.395 % to 1.023 % and increasing accuracy from 90.33 % to 99.03 %.

#### 4.2.3. Effect of backbone network architecture

Herein, we analyze the effect of different choices of CNN backbones for the extraction of depth and RGB features. Comparative results between MobileNetV3(Small), MobileNetV3(Large), and MobileNetV2 architectures (Table 6) show the MobileNetV3-Large variant offers considerably improved accuracy at a marginal computational cost, making it a preferred choice.

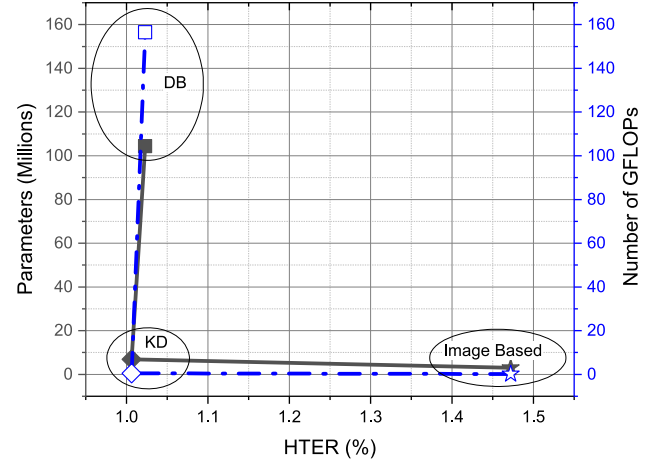


Fig. 7. FacePAD performance on the Rose-Youtu dataset vs. computational complexity in terms of number of parameters (black line) and GFLOPs (blue line). The lower values are better. An optimal FacePAD model balances performance and computational cost, as achieved by KD model.

#### 4.2.4. Effect of depth-map quality

We evaluated the impact of depth-map quality by comparing three variants of our DEN, as described in sub-section 3.3. These *Small*, *Base*, and *Large* variants are evaluated with our proposed DB model, using MobileNetV3-Large backbones. The results in Table 7 reaffirm our selection of the Base DEN variant for optimal trade-off between inference efficiency and performance. Thus, we have utilized the Base DEN variant as default for depth-map prediction in our implementations.

#### 4.3. Knowledge distillation for efficient deployment and enhanced performance

Given the computational complexity introduced by depth estimation and Dual-Branch processing, we employed Knowledge Distillation (KD). A single lightweight MobileNetV3-Large model was trained via distillation from the Dual-Branch depth-augmentation model, resulting in significant efficiency improvements while preserving accuracy. Table 8 summarizes the performance comparison, clearly demonstrating the effectiveness of the KD approach for practical real-time deployment.

Table 9 presents a detailed comparison of computational efficiency across the image-depth hybrid models including Dual-Branch model, and the final KD model. Notably, all hybrid models in this work require both the input RGB video frame and a corresponding depth-map obtained via the monocular DEN, contributing to their computational load. In contrast, the KD model eliminates this dependency, operating solely on RGB input, having learned to generalize both visual texture and geometric depth cues through the distillation process. It achieves more than 90 % reduction in FLOPs and GPU memory footprint with great reduction in inference latency while retaining strong performance.

To further illustrate the practical efficiency and effectiveness of our KD approach, Fig. 7 compares FacePAD model performance against computational complexity, specifically highlighting the number of model parameters (black line) and GFLOPs (blue line). Our KD-based model achieves an optimal balance by maintaining low HTER values while drastically reducing computational complexity.

#### 4.3.1. Knowledge distillation vs network compression

Network compression techniques such as pruning and quantization are widely used to reduce model complexity and computational overhead. For example, MobileNetV3-Small can be regarded as a pruned and optimized version of MobileNetV3-Large, obtained by systematically removing less significant channels and layers. While pruning effectively reduces parameters and memory footprint, it often compromises

**Table 4**

Results and comparison on RA, RM, RY, and OULU datasets for hybrid models with different depth-augmentation techniques.

| Datasets Model | Metric   | Channel Expansion | Auxiliary Depth | Dual-Branch (Summation) | Dual-Branch (Attention) | Dual-Branch (Concatenation) |
|----------------|----------|-------------------|-----------------|-------------------------|-------------------------|-----------------------------|
| RA             | HTER     | 0                 | 0               | 0                       | 0                       | 0                           |
|                | AUC-ROC  | 1                 | 1               | 1                       | 1                       | 1                           |
|                | Accuracy | 100               | 100             | 100                     | 100                     | 100                         |
| RM             | HTER     | 0                 | 0               | 0                       | 0                       | 0                           |
|                | AUC-ROC  | 1                 | 1               | 1                       | 1                       | 1                           |
|                | Accuracy | 100               | 100             | 100                     | 100                     | 100                         |
| RY             | HTER     | 1.395             | 0.950           | 1.780                   | 0.839                   | 1.023                       |
|                | AUC-ROC  | 0.998             | 0.998           | 0.997                   | 0.999                   | 0.999                       |
|                | Accuracy | 98.341            | 98.741          | 97.311                  | 98.627                  | 99.027                      |
| OULU-NPU       | HTER     | 0.417             | 0.729           | 0.521                   | 1.771                   | 0.313                       |
|                | AUC-ROC  | 1                 | 0.999           | 0.999                   | 0.998                   | 1                           |
|                | Accuracy | 99.390            | 99.390          | 99.222                  | 98.222                  | 99.610                      |
|                | APCER    | 0.972             | 0.417           | 1.111                   | 1.806                   | 0.278                       |
|                | BPCER    | 0.833             | 1.944           | 0.278                   | 3.611                   | 1.111                       |
|                | ACER     | 0.903             | 1.181           | 0.694                   | 2.708                   | 0.694                       |

**Table 5**

Comparison of transfer learning on ROSE-Youtu dataset.

| Metric Model       | DB V3-L (N) | DB V3-L (P) |
|--------------------|-------------|-------------|
| Test Loss          | 0.2722      | 0.0353      |
| Test Accuracy (%)  | 90.332      | 99.028      |
| AUC-ROC            | 0.9494      | 0.9986      |
| EER (%)            | 11.8553     | 1.1547      |
| HTER (%)           | 11.3954     | 1.0228      |
| FAR                | 0.0831      | 0.0115      |
| FRR                | 0.1448      | 0.0089      |
| Youden Index (Max) | 0.0308      | 0.0307      |

DB V3-L (N): Dual-Branch Model using MobileNetV3-Large backbones without pretraining, DB V3-L (P): Dual-Branch Model using MobileNetV3-Large backbones with pretraining.

**Table 6**

Comparison of depth augmented-based different MobileNet architectures on ROSE-Youtu dataset.

| Metric Model                    | DB V3-S   | DB V3-L   | DB V2     |
|---------------------------------|-----------|-----------|-----------|
| Input Resolution                | 224 × 224 | 224 × 224 | 224 × 224 |
| GFLOPs*                         | 0.1130    | 0.4444    | 0.6600    |
| Number of Parameters (M)*       | 2.4451    | 6.9282    | 5.7594    |
| Test Loss                       | 0.1019    | 0.0353    | 0.0367    |
| Test Accuracy (%)               | 97.311    | 99.026    | 98.799    |
| AUC-ROC                         | 0.9953    | 0.9986    | 0.9991    |
| EER (%)                         | 2.9253    | 1.1547    | 1.08      |
| HTER (%)                        | 2.4910    | 1.0228    | 1.2070    |
| FAR                             | 0.0231    | 0.0115    | 0.0108    |
| FRR                             | 0.0267    | 0.0089    | 0.0134    |
| Youden Index (Max)              | 0.9502    | 0.9795    | 0.9759    |
| Avg. Inference Time (s)/Sample* | 6.5491    | 7.8496    | 6.5800    |

DB V3-S: Dual-branch model using MobileNetV3-Small backbones with pretraining, DB V3-L: Dual-branch model using MobileNetV3-Large backbones with pretraining, DB V2: Dual-branch model using MobileNetV2 backbones with pretraining. \* For CNN models' combined computation of the backbone feature extraction ( $F$ ) and classification ( $C$ ) modules only, excluding the DEN.

performance, especially in tasks requiring nuanced multi-modal feature representation. This limitation is evident in our experimental results (see Section 4.2.3 and Table 6), where pruned backbones underperform relative to their larger counterparts.

Moreover, all depth-augmented architectures-including pruned variants-remain dependent on the Depth Estimation Network (DEN), which accounts for over 90 % of the total computational burden (see

**Table 7**

Comparison of depth augmented dual-branch MobileNet-V3-Large model (DB V3-L) with different DEN variants for depth-map quality on ROSE-Youtu dataset.

| Metric DEN Variant             | Base      | Small     | Large     |
|--------------------------------|-----------|-----------|-----------|
| Input Resolution (resized)     | 224 × 224 | 224 × 224 | 224 × 224 |
| Test Loss                      | 0.0353    | 0.0821    | 0.0456    |
| Test Accuracy (%)              | 99.0275   | 97.3112   | 98.7986   |
| AUC-ROC                        | 0.9986    | 0.9972    | 0.9989    |
| EER (%)                        | 1.1547    | 3.46      | 1.54      |
| HTER (%)                       | 1.0228    | 2.5213    | 1.2070    |
| FAR                            | 0.0115    | 0.0192    | 0.0108    |
| FRR                            | 0.0089    | 0.0312    | 0.0134    |
| Youden Index (Max)             | 0.9795    | 0.9496    | 0.9759    |
| Avg. Inference Time (s)/Sample | 51.3454   | 32.7988   | 136.7075  |

Base: DB V3-L model with Base DEN variant, Small: DB V3-L model with Small DEN variant, Large: DB V3-L model with Large DEN variant.

**Table 8**

Performance evaluation of the proposed method (using KD framework) on four datasets.

| Metric Datasets    | Replay-Attack | Replay-Mobile | ROSE-Youtu | OULU-NPU |
|--------------------|---------------|---------------|------------|----------|
| Test Loss          | 0.0042        | 0.0014        | 0.0504     | 0.0338   |
| Test Accuracy (%)  | 100           | 100           | 98.34      | 98.83    |
| AUC-ROC            | 1             | 1             | 0.9988     | 0.9997   |
| EER (%)            | 0             | 0             | 0.0139     | 0.0083   |
| HTER (%)           | 0             | 0             | 1.0063     | 0.7986   |
| FAR                | 0             | 0             | 0.0023     | 0.0132   |
| FRR                | 0             | 0             | 0.0178     | 0.0028   |
| Youden Index (Max) | 1             | 1             | 0.0914     | 0.9840   |

Table 9). Thus, pruning alone cannot resolve the core inefficiency associated with runtime depth computation.

In contrast, Knowledge Distillation (KD) offers a principled alternative by transferring the representational power of a complex Dual-Branch RGB + Depth model into a compact single-stream student model. The KD-based model is trained to replicate the teacher's soft predictions and intermediate feature behavior, allowing it to internalize both visual texture and geometric depth cues-without requiring depth input or estimation at inference time. As demonstrated in Table 9 and Fig. 7, our KD model achieves over 90 % reduction in FLOPs and parameters, with minimal change in accuracy, thereby making it highly suitable for real-time deployment scenarios.

**Table 9**

Efficiency comparison of different model configurations. The KD model offers a drastic reduction in FLOPs, parameters, and memory consumption while maintaining competitive performance.

| Model                       | GFLOPs |       | Params (M) |       | Model Size (MB) |       | Peak GPU Memory (MB) |       | Inference Time (ms) |       |
|-----------------------------|--------|-------|------------|-------|-----------------|-------|----------------------|-------|---------------------|-------|
|                             | DEN    | F + C | DEN        | F + C | DEN             | F + C | DEN                  | F + C | DEN                 | F + C |
| Channel Expansion           | 156.03 | 0.23  | 97.47      | 2.97  | 371.94          | 11.56 | 557.9                | 27.30 | 43.50               | 4.00  |
| Auxiliary Depth             | 156.03 | 0.29  | 97.47      | 3.50  | 371.94          | 13.57 | 557.9                | 30.12 | 43.50               | 4.21  |
| Dual-Branch (Summation)     | 156.03 | 0.47  | 97.47      | 6.44  | 371.94          | 24.97 | 557.9                | 41.45 | 43.50               | 7.75  |
| Dual-Branch (Attention)     | 156.03 | 0.47  | 97.47      | 6.93  | 371.94          | 26.84 | 557.9                | 43.91 | 43.50               | 7.74  |
| Dual-Branch (Concatenation) | 156.03 | 0.44  | 97.47      | 6.93  | 371.94          | 26.87 | 557.9                | 42.56 | 43.50               | 7.85  |
| KD-based                    | ×      | 0.22  | ×          | 3.46  | ×               | 13.42 | ×                    | 29.79 | ×                   | 3.88  |

Notation: F + C refers to the combined computation of the feature extraction and classification modules, excluding the DEN.

#### 4.3.2. Knowledge distillation vs other design choices

Our comparative analysis also evaluates alternate architectural design strategies. As shown in Tables 3, a model trained using only RGB inputs (image-only) underperforms compared to hybrid model (dual-branch) additionally incorporating depth-based information. Furthermore, we explored a Channel Expansion baseline where RGB and depth maps are concatenated as a 4-channel input and passed through a shared MobileNetV3 backbone (Table 4). While this approach benefits from multi-modal input, it is limited by its inability to separately encode modality-specific characteristics, leading to suboptimal performance.

In contrast, our Dual-Branch model (with concatenation fusion) leverages two dedicated CNN backbones—one for RGB and one for depth—allowing each modality to be independently and optimally processed. The RGB branch captures fine-grained textural information, while the depth branch focuses on geometric and structural cues. This separation avoids feature interference, leading to more robust representations and superior generalization, as evident from the results in Table 4.

Finally, our Knowledge Distillation (KD) strategy transfers the representational richness of this Dual-Branch model into a single, lightweight student model that requires only RGB input at inference. The resulting KD model achieves a favorable trade-off between accuracy and computational cost, as quantified in Table 9 and visualized in Fig. 7, validating the superiority of our design choice.

#### 4.4. Comparison with state-of-the-art methods

We comprehensively benchmarked our approach against current state-of-the-art methods across RA, RM, RY, and OULU-NPU datasets.

##### 4.4.1. Replay-attack dataset

Table 10 shows that both the Dual-Branch and KD models achieve perfect HTER scores (0.0%), outperforming recent methods such as EfficientNet-B0 [52], Multi-Block LBP [53], and MobileNet + Image Diffusion [54].

##### 4.4.2. Replay-mobile dataset

The results detailed in Table 11 further affirm the superior performance of our approach, maintaining ideal HTER performance (0.0%) for both Dual-Branch and KD models, compared to recent methods like EfficientNet-B0 [52] and Lightweight 3D-DNN [61].

##### 4.4.3. ROSE-Youtu dataset

Comparative results on the ROSE-Youtu dataset are summarized in Table 12. The proposed method achieves outstanding performance, notably reducing HTER to as low as 1.006% for KD model and 1.023% for Dual-Branch model.

##### 4.4.4. OULU-NPU dataset

Table 13 presents a comprehensive comparison of performance on the OULU-NPU dataset, highlighting the effectiveness of our proposed Dual-Branch (DB) and knowledge distillation (KD) models utilizing both MobileNetV3-Large and MobileNetV2 backbones.

**Table 10**

Comparative test results for replay-attack dataset.

| Method                                     | Year        | HTER (%)   |
|--------------------------------------------|-------------|------------|
| InceptionV4 [55]                           | 2020        | 13.54      |
| SCNN [55]                                  | 2020        | 7.53       |
| SfSNet [56]                                | 2020        | 3.1        |
| VGG16 + GMM [57]                           | 2021        | 1.46       |
| GoogLeNet + GMM [57]                       | 2021        | 3.76       |
| MIQF + SVM [58]                            | 2022        | 5.38       |
| 3D ConvNet [59]                            | 2023        | 11.7       |
| Multi-Block LBP [53]                       | 2023        | 6.98       |
| MobileNet + Image Diffusion [54]           | 2023        | 0.09       |
| ResNet50V2 [60]                            | 2023        | 0.03       |
| Lightweight 3D-DNN [61]                    | 2024        | 0.0        |
| HaTFAS [62]                                | 2024        | 0.0        |
| EfficientNet-B0 [52]                       | 2024        | 36.88      |
| HybridNet II [63]                          | 2025        | 0.02       |
| HybridNet I [63]                           | 2025        | 0.0        |
| Deformable Convolution [49]                | 2025        | 0.0        |
| Spatio-Temporal [41]                       | 2025        | 0.0        |
| AdvSpoofGuard [48]                         | 2025        | 0.0        |
| <b>Proposed Method (Dual-Branch Model)</b> | <b>2025</b> | <b>0.0</b> |
| <b>Proposed Method (KD Model)</b>          | <b>2025</b> | <b>0.0</b> |

**Table 11**

Comparative test results for replay-mobile dataset.

| Method                                     | Year        | HTER (%)   |
|--------------------------------------------|-------------|------------|
| SMKFNS [64]                                | 2020        | 11.88      |
| InceptionV4 [55]                           | 2020        | 5.94       |
| SCNN [55]                                  | 2020        | 4.96       |
| VGG16 + GMM [57]                           | 2021        | 17.21      |
| GoogLeNet + GMM [57]                       | 2021        | 13.56      |
| MK-SVDD-Slim [65]                          | 2021        | 7.6        |
| MKL [65]                                   | 2021        | 6.7        |
| WA (PSO + PS) [66]                         | 2021        | 5.85       |
| WA (GA + MMS + PS) [67]                    | 2022        | 5.35       |
| 3D ConvNet [59]                            | 2023        | 8.7        |
| MobileNet + Image Diffusion [54]           | 2023        | 1.14       |
| ResNet50V2 [60]                            | 2023        | 0.0        |
| EfficientNet-B0 [52]                       | 2024        | 4.62       |
| Lightweight 3D-DNN [61]                    | 2024        | 0.45       |
| Deformable Convolution [49]                | 2025        | 0.0        |
| Spatio-Temporal [41]                       | 2025        | 0.0        |
| AdvSpoofGuard [48]                         | 2025        | 0.0        |
| <b>Proposed Method (Dual-Branch Model)</b> | <b>2025</b> | <b>0.0</b> |
| <b>Proposed Method (KD Model)</b>          | <b>2025</b> | <b>0.0</b> |

The *Dual-Branch Model (DB)* achieves an ACER of **0.69%** with MobileNetV3-Large and a notably lower ACER of **0.56%** with MobileNetV2. This strong performance demonstrates the benefit of modality-specific feature learning in distinguishing real from spoofed presentations, and confirms that even with the small sized model like MobileNetV2 architecture is highly competitive in this setting.

The *Knowledge Distillation (KD) Model* further enhances efficiency by distilling multimodal knowledge from the Dual-Branch teacher into a compact, single-branch student model. The MobileNetV3-Large KD

**Table 12**  
Comparative test results for ROSE-Youtu dataset.

| Method                                     | Year        | HTER/EER (%)       |
|--------------------------------------------|-------------|--------------------|
| AlexNet [68]                               | 2020        | -/8.0              |
| SE-ResNet-18 [69]                          | 2020        | -/8.0              |
| ResNet50 + GMM [57]                        | 2021        | 14.69/-            |
| De-Spoofing [70]                           | 2021        | -/12.3             |
| RCTR-All Spaces [70]                       | 2021        | -/10.7             |
| FASNet [71]                                | 2021        | 8.57/-             |
| Color [70]                                 | 2021        | -/13.9             |
| WA (PSO + PS) [66]                         | 2021        | 5.61/-             |
| WA (GA + MMS + PS) [67]                    | 2022        | 5.12/-             |
| Fatemifar et al. [72]                      | 2022        | 6.34/-             |
| 3D ConvNet [59]                            | 2023        | 21.3/-             |
| MobileNet + Image Diffusion [54]           | 2023        | 4.92/4.95          |
| ResNet50V2 [60]                            | 2023        | 2.53/2.64          |
| ViViT [73]                                 | 2023        | 13.28/2.46         |
| EfficientNet-B0 [52]                       | 2024        | 9.54/-             |
| CA-FAS [74]                                | 2024        | 1.37/-             |
| AdvSpoofGuard [48]                         | 2025        | 1.97/1.08          |
| Spatio-Temporal ( $\tau = 1$ ) [41]        | 2025        | 1.472/0.85         |
| Deformable Convolution [49]                | 2025        | 1.26/0.80          |
| <b>Proposed Method (Dual-Branch Model)</b> | <b>2025</b> | <b>1.023/1.15</b>  |
| <b>Proposed Method (KD Model)</b>          | <b>2025</b> | <b>1.0063/1.39</b> |

**Table 13**  
Comparative test results for OULU-NPU complete dataset.

| Method                  | Year        | APCER (%)   | BPCER (%)   | ACER (%)    |
|-------------------------|-------------|-------------|-------------|-------------|
| CS-LBP [75]             | 2020        | 14.9        | 10.5        | 12.7        |
| CS-VGG [75]             | 2020        | 3.3         | 9.4         | 6.35        |
| ED-LBP (VAR) [76]       | 2021        | 11.3        | 8.4         | 9.9         |
| Texture (VAR) [77]      | 2021        | 14.5        | 15          | 14.8        |
| Fake-Net (VAR) [78]     | 2021        | 5.4         | 6.9         | 6.2         |
| OFT (VAR) [79]          | 2022        | 5.7         | 2.7         | 4.2         |
| ELA [80]                | 2023        | –           | –           | 4.86        |
| 3D-LCN [81]             | 2024        | 1.5         | 0.5         | 1.0         |
| Spatio-Temporal [41] ‡  | 2025        | 1.94        | 0.55        | 1.25        |
| <b>DB (MobileNetV3)</b> | <b>2025</b> | <b>0.28</b> | <b>1.11</b> | <b>0.69</b> |
| <b>KD (MobileNetV3)</b> | <b>2025</b> | <b>1.39</b> | <b>0.83</b> | <b>1.11</b> |
| <b>DB (MobileNetV2)</b> | <b>2025</b> | <b>0.28</b> | <b>0.83</b> | <b>0.56</b> |
| <b>KD (MobileNetV2)</b> | <b>2025</b> | <b>0.28</b> | <b>0.83</b> | <b>0.56</b> |

Proposed (DB: Dual-Branch, KD: Knowledge-Distillation), CS: Compact Space, VAR: Video Attack Return, OFT: Optical Flow + Texture, ‡ means the method is retested on this dataset.

variant achieves an ACER of **1.11 %**, while the MobileNetV2 KD variant maintains the impressive ACER of **0.56 %**, matching the performance of its Dual-Branch counterpart. This highlights that knowledge distillation can produce highly efficient models for real-time and resource-constrained applications without significant loss in detection accuracy.

When benchmarked against traditional and recent state-of-the-art methods, both the DB and KD models (for both MobileNetV3-Large and MobileNetV2) deliver superior results, positioning them among the top-performing approaches in the literature. Only a handful of highly specialized methods report marginally better ACER, often with greater architectural or computational complexity.

#### Key observations:

- Both MobileNetV3-Large and MobileNetV2 are effective as backbone networks for the proposed Dual-Branch architecture, with MobileNetV2 achieving the lowest ACER, offering a strong balance between accuracy and efficiency.
- The knowledge distillation framework enables the deployment of a lightweight, single-branch student network with minimal loss in performance, removing the need for explicit depth estimation at inference.
- Across all variants, our method outperforms most prior works, demonstrating strong generalization, robustness, and suitability for real-world deployment.

Overall, the proposed Dual-Branch and knowledge-distilled models provide a robust and efficient FacePAD solution, consistently delivering state-of-the-art results with minimal computational overhead, making them well suited for mobile authentication, surveillance, and other embedded biometric security systems.

#### 4.4.5. Summary of comparative results

Overall, the proposed Dual-Branch model and its distilled variant significantly surpass state-of-the-art methods across all benchmark datasets evaluated. The Dual-Branch approach's integration of RGB and depth modalities, combined with effective knowledge distillation, results in a powerful yet lightweight and efficient FacePAD solution, suitable for real-time, resource-constrained applications.

#### 4.5. Potential failure cases and limitations

We conducted a focused failure case analysis on the ROSE-Youtu test set to evaluate the robustness of our proposed approach and to identify potential limitations. The dataset comprises a diverse set of challenging spoofing scenarios, including: (i) static and quivering printed paper attacks, (ii) video replay attacks displayed on various screen types, and (iii) paper mask attacks in uncut, cropped, and partially cropped formats. This diversity simulates real-world presentation attack conditions with variations in lighting, motion artifacts, and occlusions.

Despite the complexity of these scenarios, our Dual-Branch and Knowledge Distillation models exhibited strong generalization and reliability, achieving *specificity* scores of *0.9977* and *0.9985*, respectively. Only 3 and 2 attack samples were misclassified across the entire test set, underscoring the models' high capability to reject unauthorized access—a crucial requirement for real-world face authentication systems.

Nevertheless, occasional misclassifications occurred in borderline cases—particularly where the estimated depth map was imperfect—such as in scenes involving partial facial occlusion, severe lighting glare, or motion-induced blur. These artifacts often compromised the quality of extracted depth or texture cues, thereby affecting the model's decision confidence. Furthermore, as access to affordable and high-fidelity 3D printing improves, realistic 3D facial masks may emerge as a more serious challenge requiring enhanced geometric modeling or anti-spoofing depth consistency checks.

Lastly, while not yet prevalent in consumer-level attacks, digital adversarial perturbations remain a notable vulnerability. As highlighted in our earlier work [48], such attacks—although synthetic in origin—can expose model weaknesses and may be relevant in high-security contexts. These limitations suggest several promising directions for future work, including the incorporation of temporal attention mechanisms, adaptive preprocessing pipelines, adversarial training strategies, and additional multimodal fusion beyond RGB and depth to improve resilience under extreme or adversarial conditions.

## 5. Conclusion

In this paper, we introduced a robust and efficient face presentation attack detection (FacePAD) framework that combines both RGB images and synthetically generated depth-maps using a Dual-Branch architecture. By leveraging complementary information from texture and depth cues, our method significantly improves spoof detection accuracy and generalization across diverse attack scenarios and benchmark datasets.

We further proposed a knowledge distillation scheme that transfers the rich multimodal knowledge from the Dual-Branch model to a lightweight, single-branch MobileNetV3 or MobileNetV2 backbone. This approach enables practical deployment on resource-constrained devices, maintaining high performance without the overhead of explicit depth estimation at inference.

Our results demonstrate that the proposed models achieve state-of-the-art performance on several challenging FacePAD datasets, including OULU-NPU, Replay-Attack, Replay-Mobile, and ROSE-Youtu. Both the

Dual-Branch and knowledge-distilled models outperform most existing methods, striking an excellent balance between accuracy and computational efficiency.

Looking ahead, future work will explore dynamic fusion strategies for adaptively weighing RGB and depth cues, incremental learning for handling new types of spoofing attacks, and extending the framework to other biometric modalities for even greater robustness in multimodal authentication systems.

### CRedit authorship contribution statement

**Muhammad Shahid Jabbar:** Investigation, Visualization, Resources, Methodology, Data curation, Validation, Software, Writing – original draft, Writing – review & editing; **Taha Hasan Masood Siddique:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Investigation, Data curation; **Kejie Huang:** Writing – review & editing, Writing – original draft, Validation, Supervision; **Shujaat Khan:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

### Data availability

The authors do not have permission to share data.

### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. The authors declare the following financial interests/personal relationships which may be considered as potential competing interests.

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